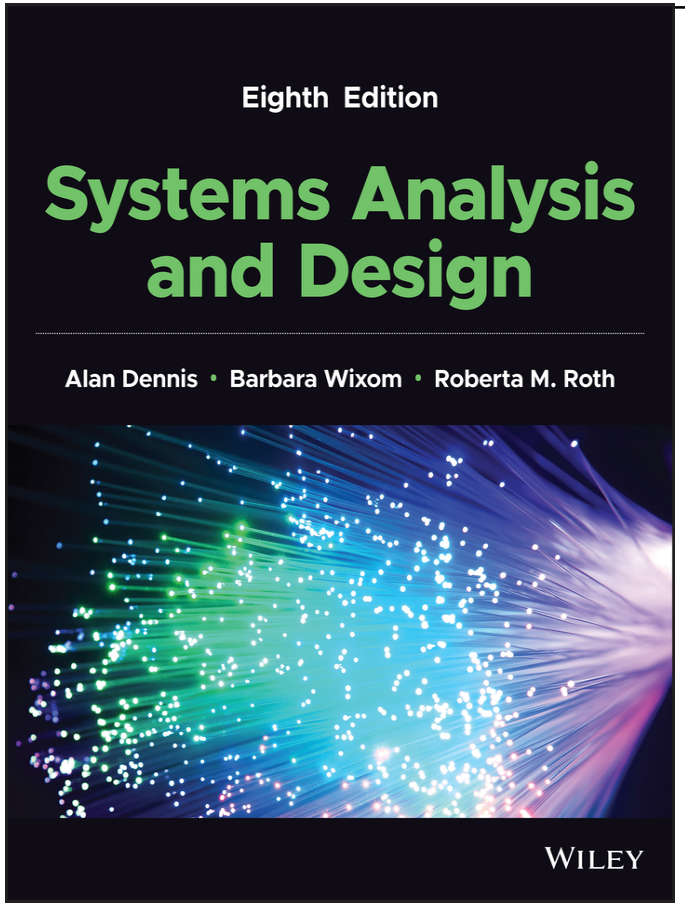
Software Development Lifecycles

Part 4



Systems Analysis and Design

Chapter 13

Agile Development Methods

Outline

- Origins of Agile
- Evolution of Agile Development
- Scrum
- Other Types of Agile Methodologies
- Comparing the SDLC with Agile Methodologies

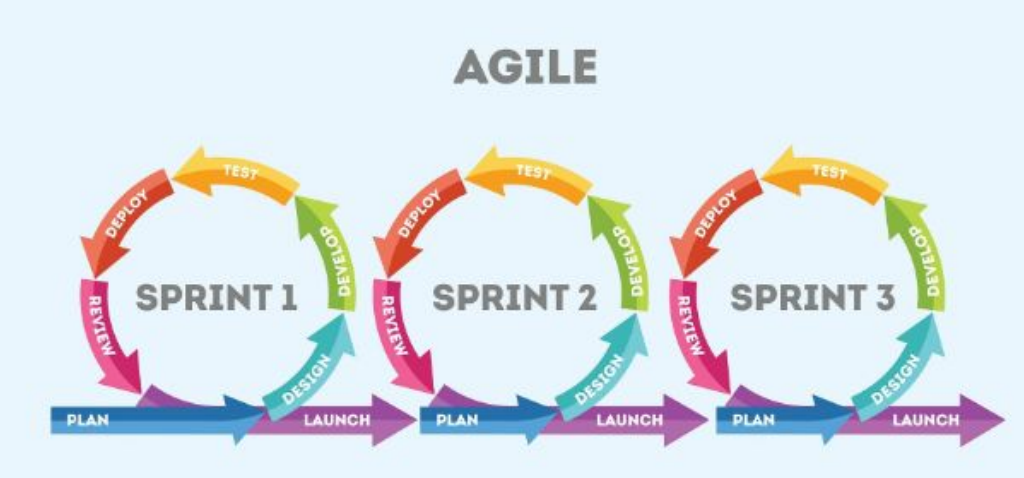
Introduction

- ***We are uncovering better ways of developing software by doing it and helping others do it.*** —The Agile Manifesto

- Discuss the foundation values and principles expressed by the Agile Manifesto.
- Look at the use of specific Agile methodologies today.
- Discuss Scrum's characteristics, features, and unique processes in detail.
- Other Agile methodologies

Origins of Agile

- A philosophy or a framework for quickly adapting and responding to change
- The desire to achieve success in a constantly volatile and uncertain environment.
- Agile philosophy is described by 4 core values and 12 core principles



Agile Manifesto values.

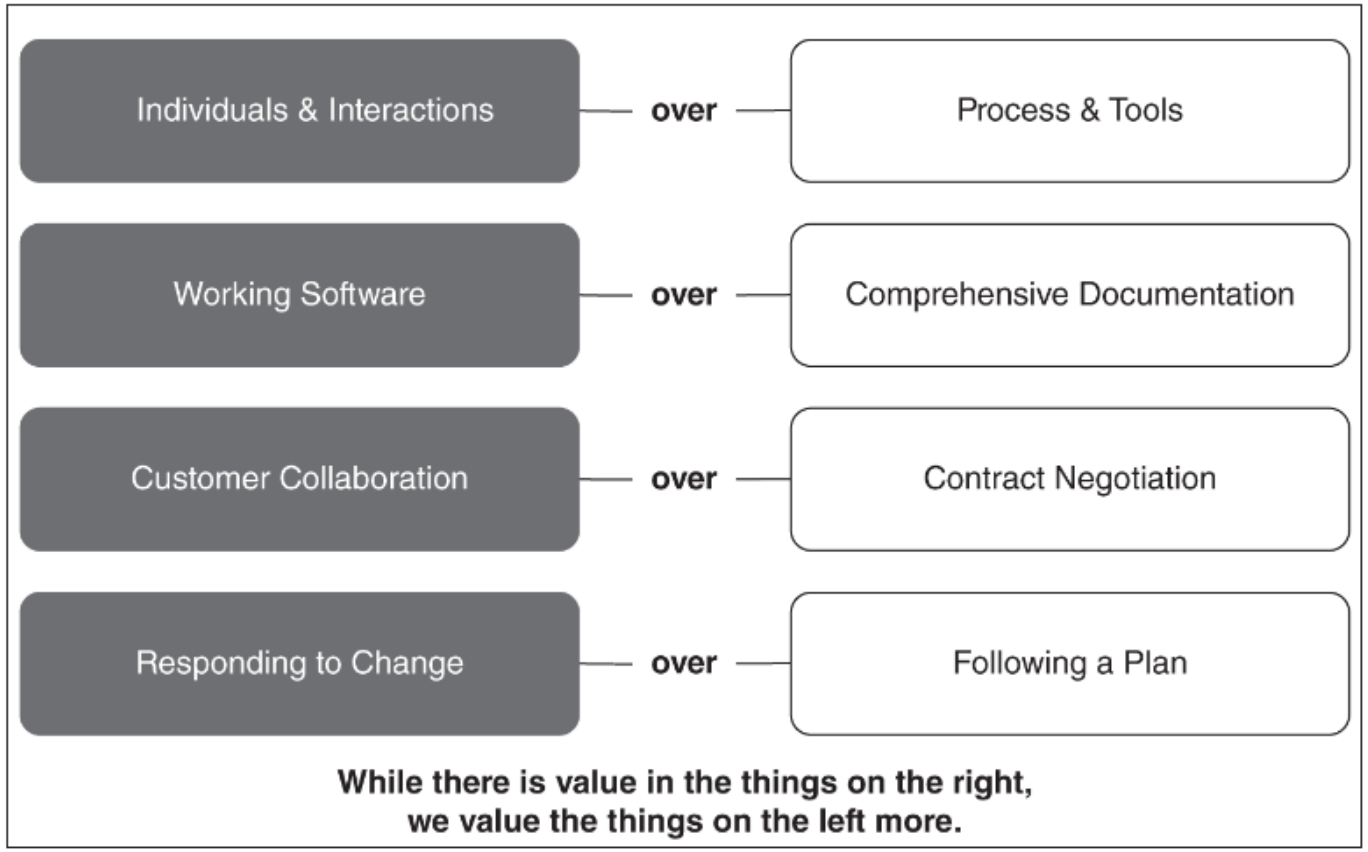


FIGURE 13-1 Agile Manifesto values.

Agile Manifesto principles.

1. Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
2. Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
3. Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
4. Businesspeople and developers must work together daily throughout the project.

Agile Manifesto principles.

5. Build projects around motivated individuals. Give them the environment and support they need and trust them to get the job done.
6. The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
7. Working software is the primary measure of progress.
8. Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Agile Manifesto principles.

9. Continuous attention to technical excellence and good design enhances agility.
10. Simplicity—the art of maximizing the amount of work not done—is essential.
11. The best architectures, requirements, and designs emerge from self-organizing teams.
12. At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

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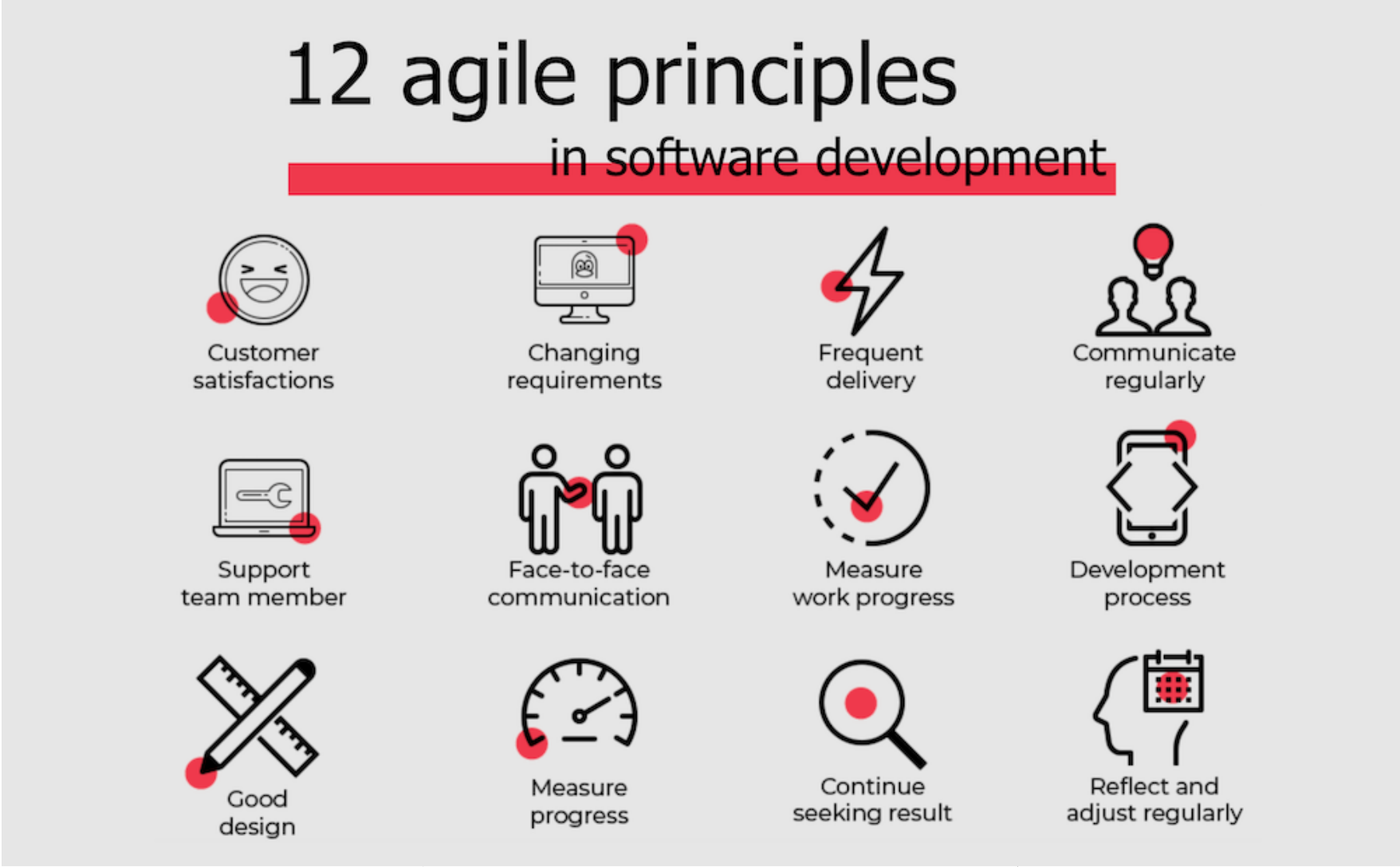
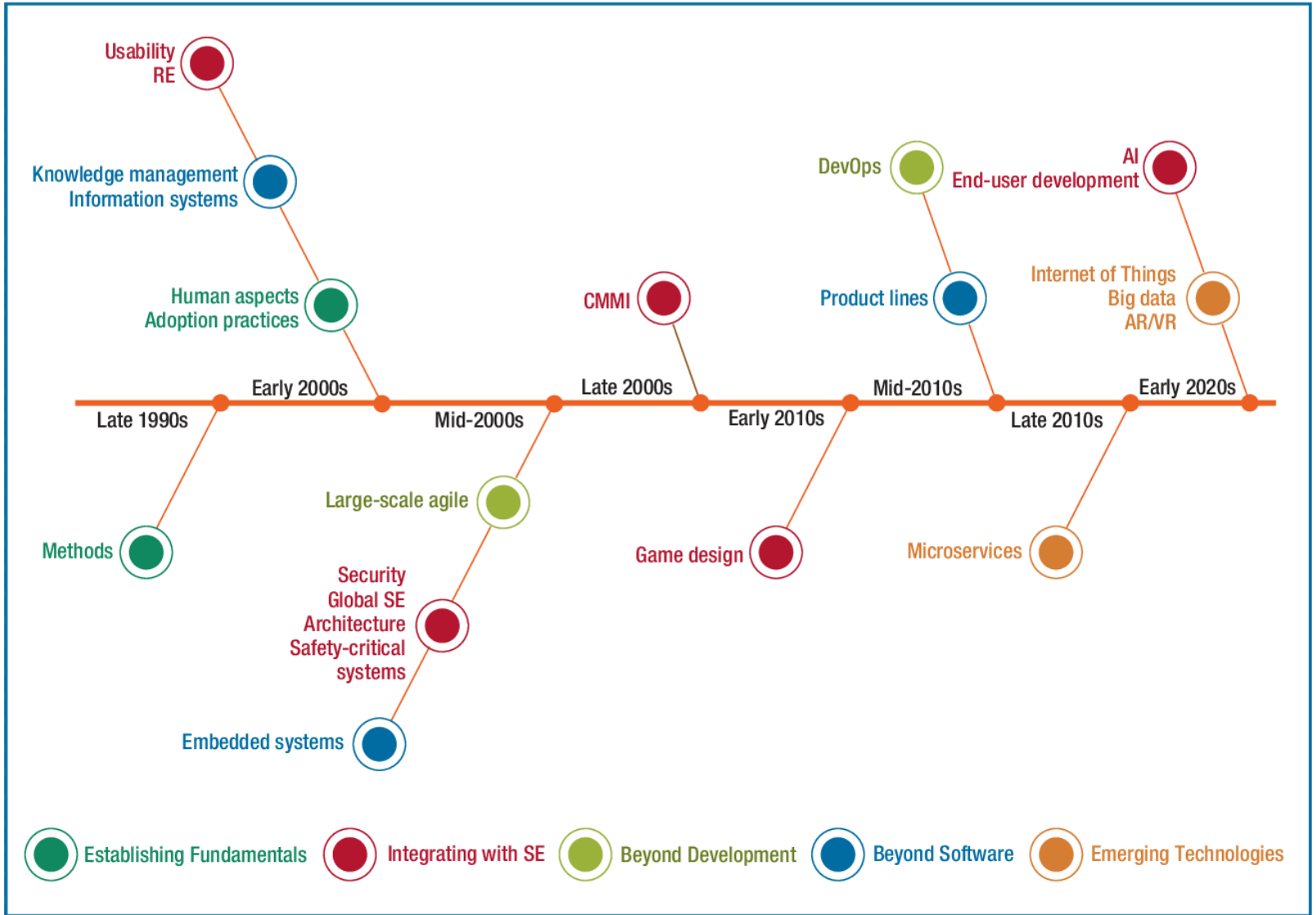

12 agile principles
in software development
Customer satisfactions
Changing requirements
Frequent delivery
Communicate regularly
Support team member
Face-to-face communication
Measure work progress
Development process
Good design
Measure progress
Continue seeking result
Reflect and adjust regularly

Summary of Agile characteristics

- Close collaboration between the project team and business experts
- Face-to-face communication
- Frequent delivery of new, deployable business value
- Tight, self-organizing teams
- Reduced impact of changes in requirements

Evolution of Agile Development



Adoption of the Agile Approach

- According to the 2020 survey, only a quarter of respondents work in companies that have been practicing Agile development for 5 years or more.
- Agile methodology mainly applies SE and IT.
- The predominant reasons for adopting Agile were to accelerate software delivery and to enhance the ability to manage changing priorities

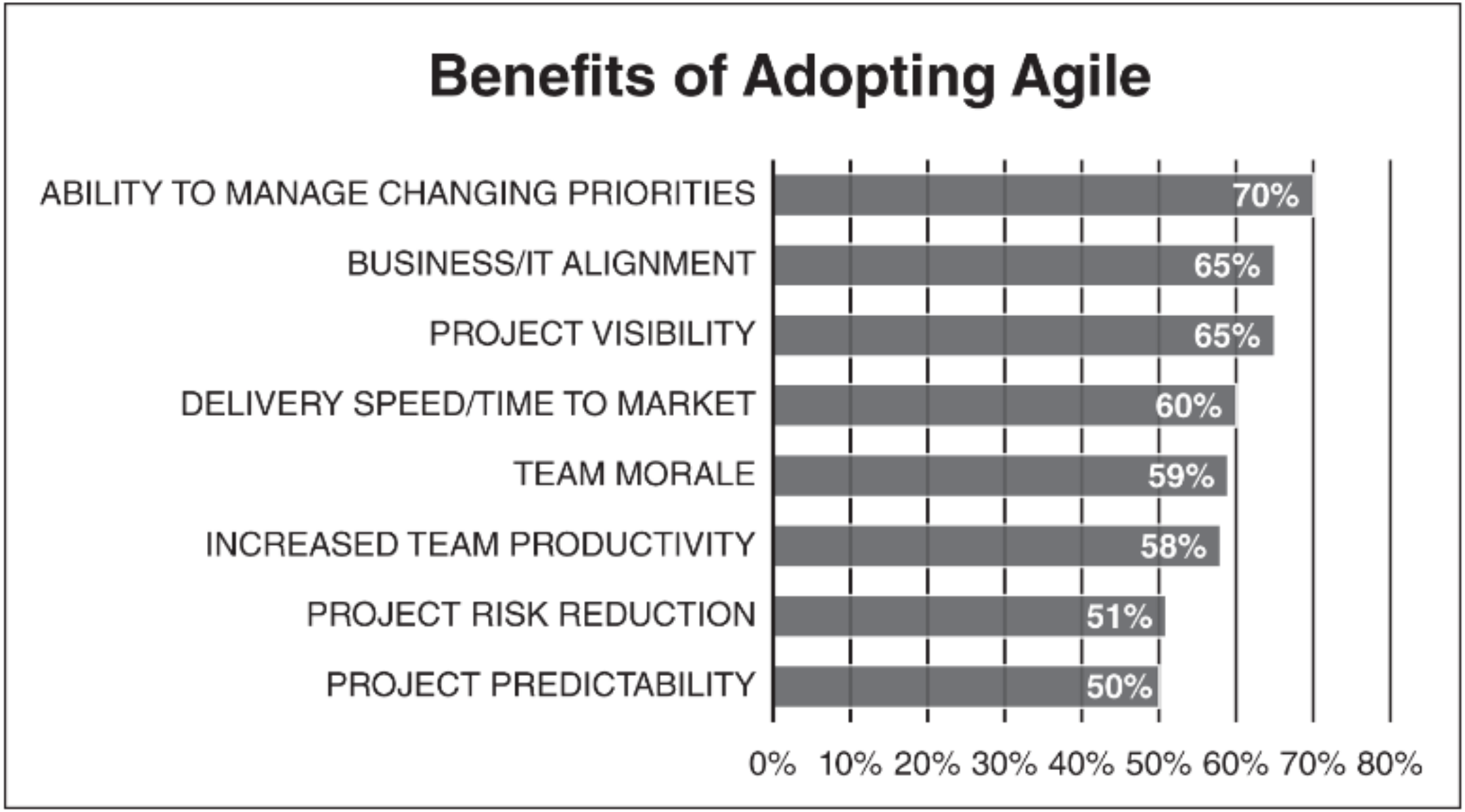


FIGURE 13-3 Benefits of adopting Agile.

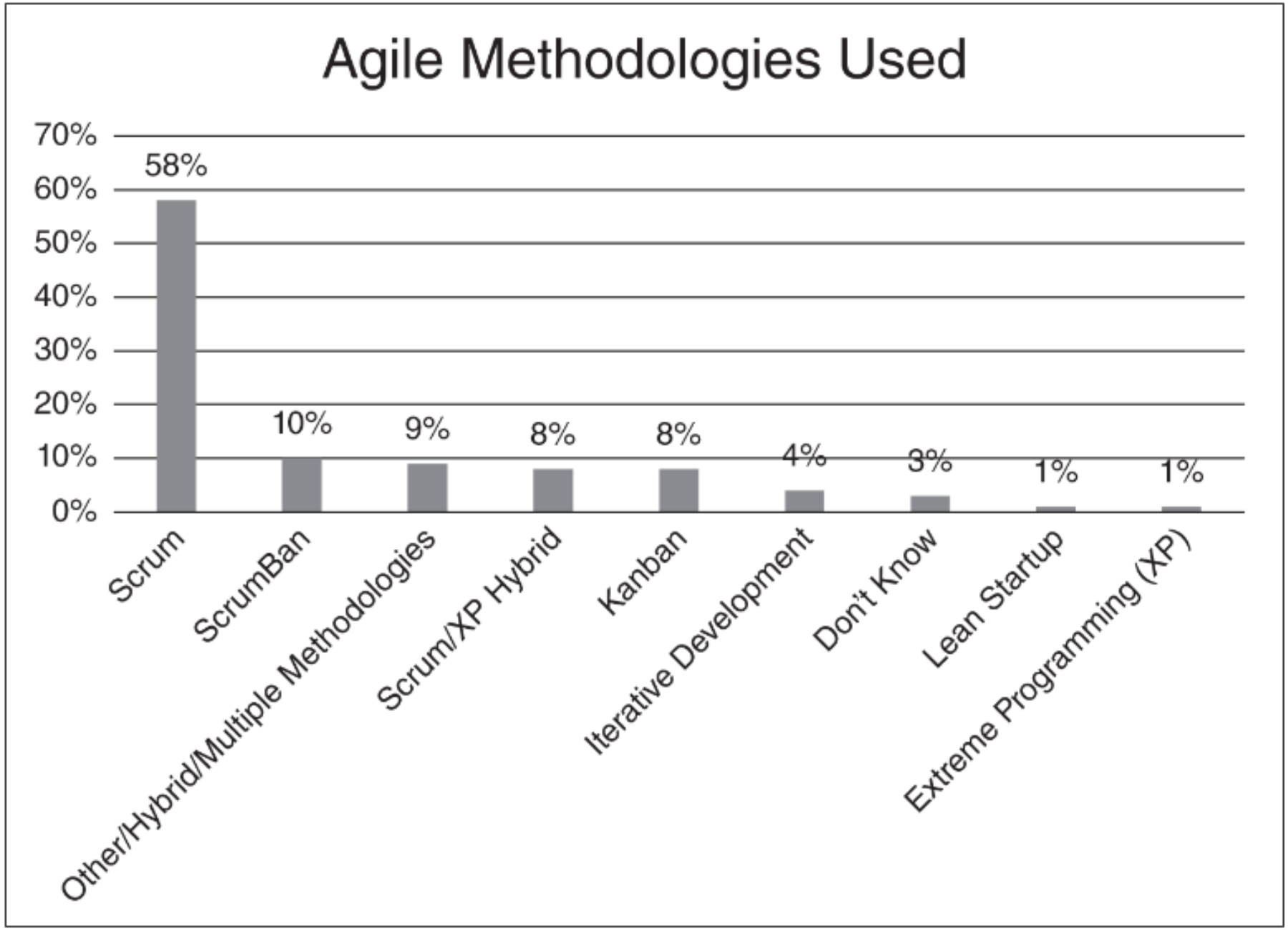
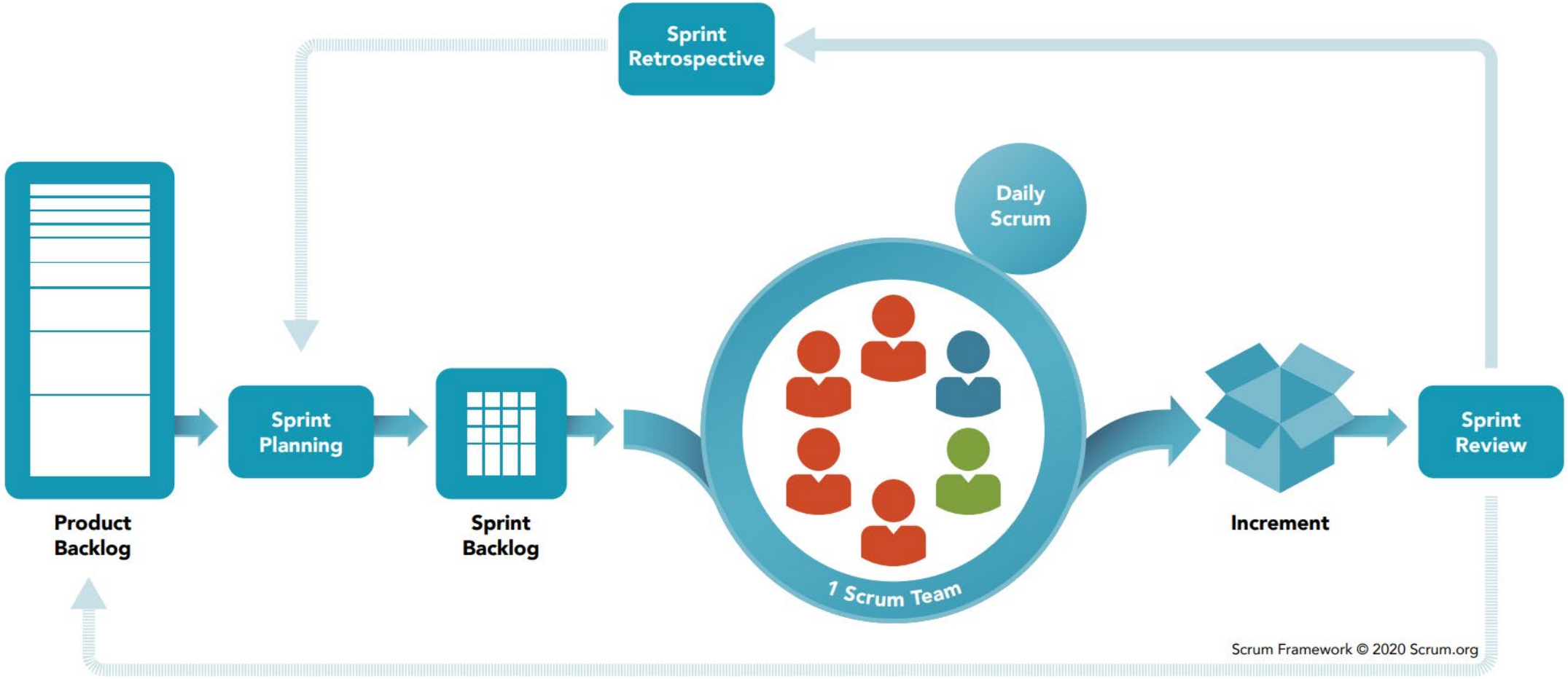


FIGURE 13-4 Agile methods in use.

Scrum


Sprint Retrospective
Daily Scrum
Product Backlog
Sprint Planning
Sprint Backlog
1 Scrum Team
Increment
Sprint Review
Scrum Framework © 2020 Scrum.org

Overview of Scrum

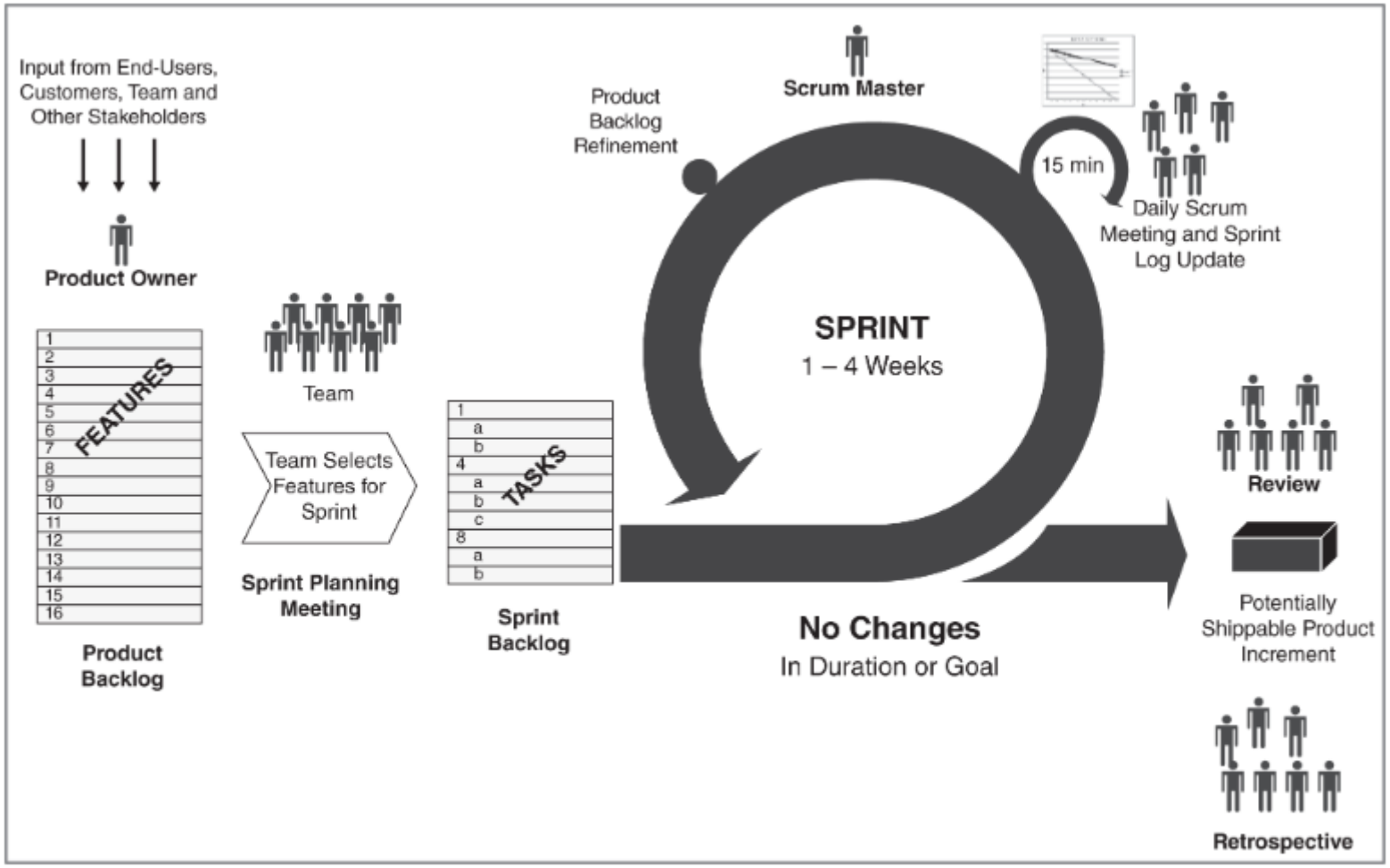


FIGURE 13-5 Scrum overview.

(cont'd)

- These items are gathered and managed by the **product owner**, who represents the organization's interests in this project.
- The product owner develops and manages a prioritized feature list, also called a **product backlog**, that serves as the development teams' to-do list.
- The development cycles in the Scrum development process are called **sprints**.
- At the beginning of a sprint, each development team assigned to the project conducts a **sprint planning meeting.**

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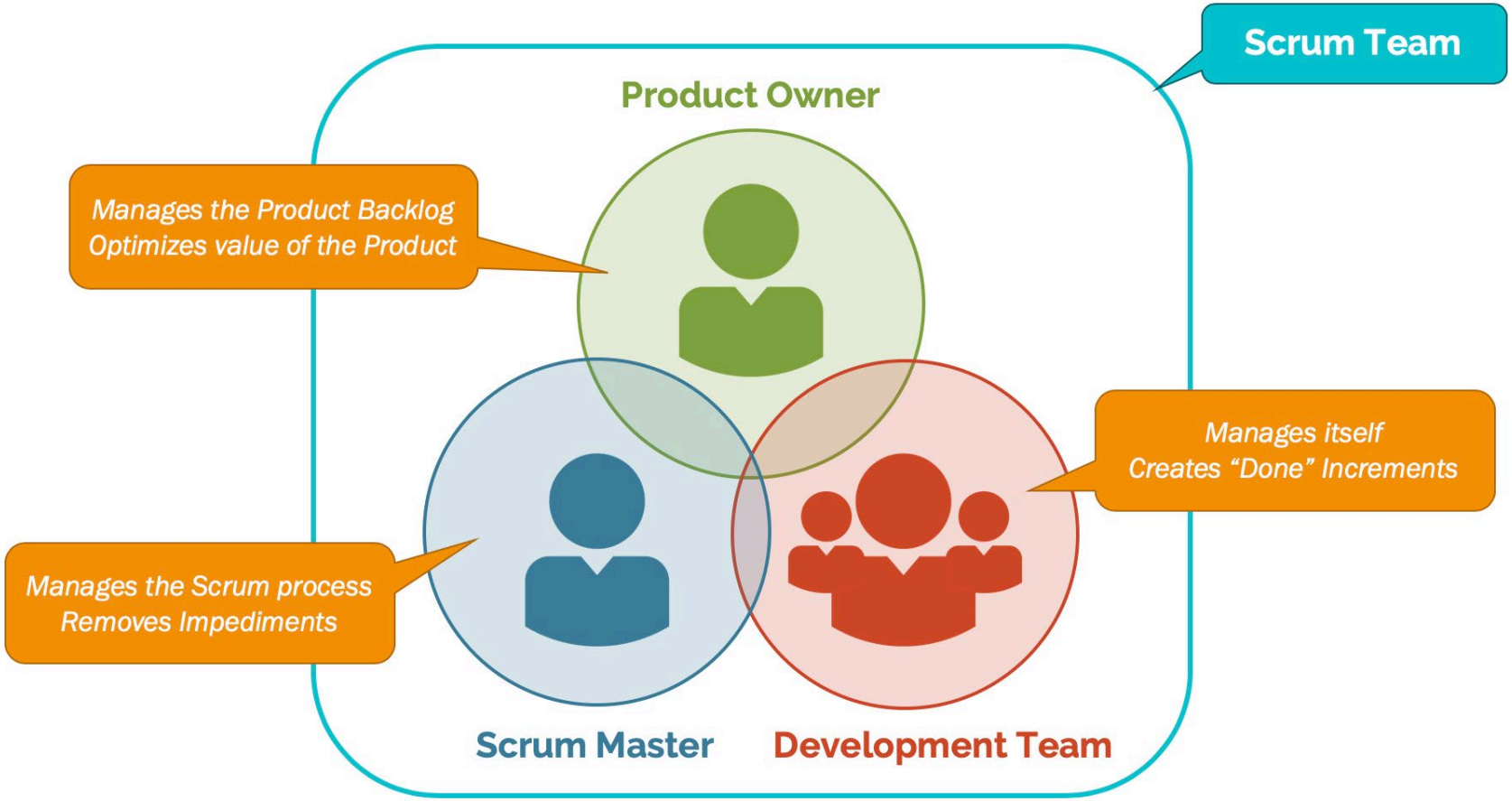
- The feature(s) are then refined into a more detailed set of tasks, called the **sprint backlog.**
- A standard feature of Scrum is the daily scrum meeting, called a **daily standup**
- This brief meeting is attended by all team members, a team member called the **ScrumMaster**
- As a final aspect of the sprint, the team performs a **sprint retrospective** on its performance in the just-completed sprint

Scrum Characteristics

- The use of dedicated, self-organizing teams
- Software products are completed in a series of short work cycles
- The system's requirements are captured from end-users, customers, and other interested stakeholders.
- No specific software engineering practices are prescribed in the Scrum methodology.

Scrum Roles


Scrum Team
Product Owner
Manages the Product Backlog
Optimizes value of the Product
Manages itself
Creates "Done" Increments
Manages the Scrum process
Removes Impediments
Scrum Master
Development Team

Product Owner

- Defines the features of the product
- Prioritizes features according to business value
- Adjusts features and priorities every sprint, as needed
- Accepts or reject work results
- Decides when to release and what's included
- Responsible for the profitability of the product (ROI)

ScrumMaster

- Provides management of the process
- Responsible for enacting Scrum values and practices
- Removes impediments
- Ensures that the team is fully functional and productive
- Enables dose cooperation across all roles and functions
- Shields the team from external. interferences

Development Team

- Typically 5-9 people
- Cross-functional: Programmers, testers, user experience, designers, etc.
- Members should be full-time
- May be exceptions (e.g., database administrator)
  - "External team members"

Development Team

- Teams are self-organizing
- Ideally no titles, but rarely a possibility
- Membership should change only between sprints

Scrum Features

- Sprints
- User Stories
- Acceptance Criteria
- Story Points
- Team Velocity

Scrum Processes

- Sprint Planning

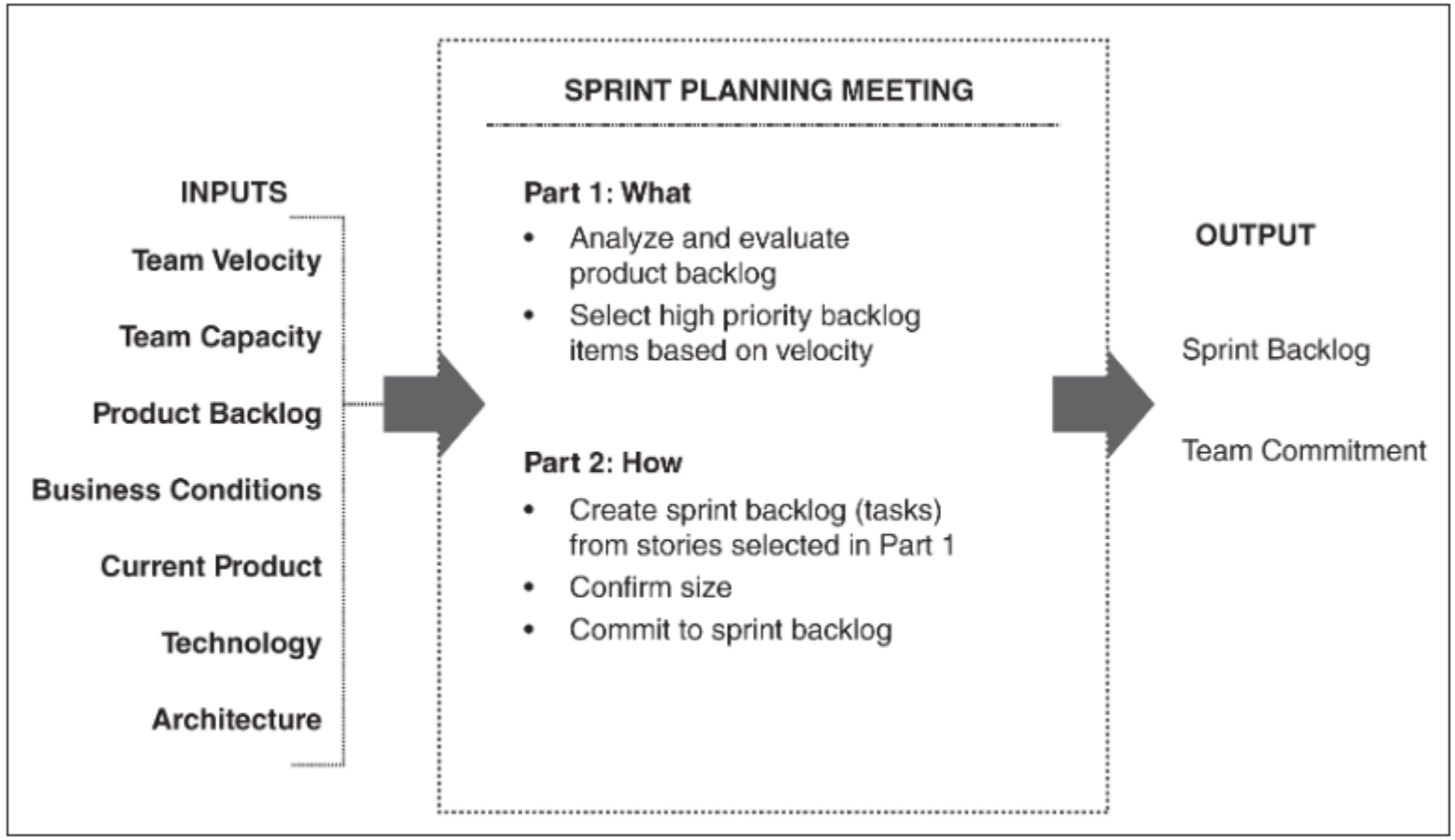


FIGURE 13-10 Sprint planning process.

Scrum Processes

- Product Backlog Grooming

Velocity—20 points

Product Backlog	
Cancels	5pts
Returns	8pts
Gift Wrap	3pts
Coupons	2pts
Wish List	13pts
Product Reviews	8pts
Tracking	20pts
Recommendations	5pts

Sprint Commitment	
Cancels	5pts
Returns	8pts
Gift Wrap	3pts
Coupons	2pts

Sprint Backlog	
Cancels	• Code the User Interface • Create cancel transaction • Add confirmation e-mail • Update Inventory
Returns	• Code the User Interface • Create Return Transaction • Add Confirmation E-mail • Update Inventory
Gift Wrap	• Allow Gift Wrap Selection • Add to Invoice • Notify Gift Wrap Group
Coupons	• Allow Coupon Entry • Update Invoice

FIGURE 13-11 Sprint planning illustration.

Scrum Processes

- Scrum Meetings

Session	Purpose	Timing/Duration	Participants
Release Planning	Determine what a release should include and when it should be delivered	Start of release 2–4 hours	Product owner, ScrumMaster, key stakeholders, architect, team (optional)
Sprint Planning	Elaborate, estimate, and prioritize highest-value product backlog items for a sprint	Start of each sprint 2–4 hours	Team, ScrumMaster, product owner
Daily Standup	Facilitate rapid coordination between team member and product owner	Daily 15 minutes	Team, ScrumMaster, product owner
Sprint Review (or Demo)	Demonstrate completed functionality to interested stakeholders and users to show progress and get feedback	End of each sprint 1–1½ hours	Team, ScrumMaster, product owner, interested stakeholders and users
Sprint Retrospective	Reflect on project and process issues within team and act as appropriate	End of each sprint 30–45 minutes	Team, ScrumMaster, product owner
Product Backlog Grooming	Review upcoming user stories to confirm size and clarify team questions and decompose to execution level	Each sprint 1–2 hours	Team, ScrumMaster, product owner

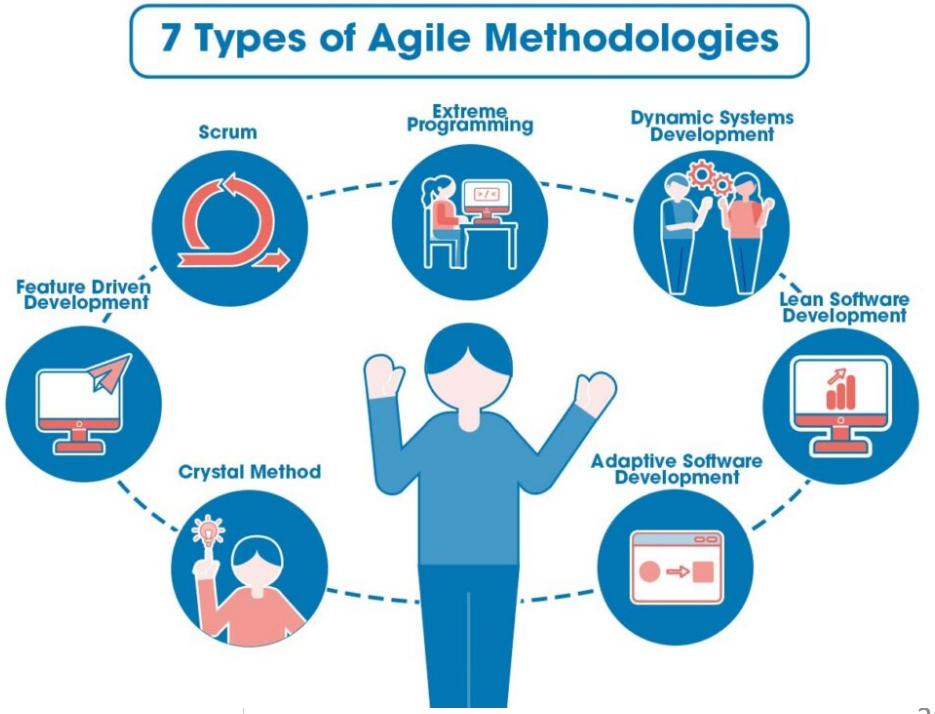
FIGURE 13-12 Scrum meetings.

How Does Scrum End?

- The product owner determines that the remaining features in the product backlog have little value.
- The project sometimes must be terminated because the project budget is exhausted
- The organization determines that the team needs to be reassigned to a different, higher value project and the project is ended.

Other Types of Agile Methodologies

- Crystal Development Methodology
- Dynamic Systems Development Methodology
- Feature Driven Development
- Lean Software Development
- …



Comparing the SDLC with Agile Methodologies

Waterfall SDLC	Agile
• Carefully planned first version	• Speed and adaptability
• Limited transparency	• Full transparency between members
• Top-down responsibility	• Shared responsibility and ownership
• Initial research and planning	• Ongoing research, planning, and testing
• Different teams in different stages	• Ongoing collaboration
• Focused on delivering full product	• Iteration-focused